

Production functional realisation audit – variational closure scaffold

TECT verification-first repository · claim A1-PRODUCTION-FUNCTIONAL-REALISATION

first issued 2026-07-17 · this version issued 2026-07-17 · v1.0

1. Purpose and scope

This record freezes and audits the current three-component complex working-branch functional on a periodic three-torus. It does not enlarge the pure-Brazovskii scalar-slice T5 claim. The authority is the external backend and configuration pinned by SHA-256 in `production_functional_manifest.json`. The audit checks whether the declared energy, executable residual, and executable Hessian are mutually compatible under the stated real inner product.

2. Frozen setting

For $\Psi \in \mathbb{C}^3$ on a spectral grid on $\mathbb{T}^3$, use

$$\langle u, v \rangle_{\mathbb{R}} = dV \operatorname{Re} \sum_{a,x} \bar{u}_a(x) v_a(x).$$

The frozen declared energy contains the Brazovskii quadratic term, family mass, internal projector lock $P_0 = z_0 z_0^\dagger / (z_0^\dagger z_0)$, shell bias, quartic and sextic density terms, and the Class-II J, K terms. The manifest gives the complete formulas, all coefficients, $\rho + 10^{-12}$ regularisation, and the separate solver floor $\rho \geq \rho_*$.

3. Audit procedure

The standalone verifier independently reconstructs the declared energy with Torch autodiff and compares it with finite differences. It then imports the hash-pinned external residual and Hessian and evaluates

$$DF[\Psi](v) = \langle R[\Psi], v \rangle_{\mathbb{R}}, \quad DR[\Psi]v = H[\Psi]v, \quad \langle u, Hv \rangle_{\mathbb{R}} = \langle Hu, v \rangle_{\mathbb{R}}.$$

It covers zero, homogeneous, random, commensurate q_0 -shell, and Class-II-active fields at three finite-difference steps on $N = 4$.

4. Current result and named obstruction

The audit is designed to pass an independent energy-derivative control and to expose the current working-branch mismatch. Already in the scalar core, $(\lambda/2)\rho^2 + (\gamma/3)\rho^3$ has real gradient $2\lambda\rho\Psi + 2\gamma\rho^2\Psi$, while the residual uses half those nonlinear terms. Independently, the declared Class-II energy uses c_{JJ} and c_{KK} , whereas the residual uses c_{JJ} and c_{JK} ; the pinned values $c_{JK} = 0.1$ and $c_{KK} = 0.15$ are unequal. Therefore the full production configuration does not currently establish a single variational functional. This is a diagnosis, not a rejection of the scalar-slice kernel result.

5. Devil’s-advocate

(alpha) “*This is only a finite-difference artifact.*” DISMISSED for the diagnosis only if the independent autodiff derivative and multi-step finite differences agree while the imported residual disagrees; all are reported.

(beta) “*Complex fields may use a different factor-of-two convention.*” DISMISSED only relative to the explicitly frozen real pairing: both the autodiff conversion and the finite-difference directional derivative use that same convention.

(gamma) “*The scalar T5 manifest has been invalidated.*” VALID only as a scope warning, with mitigation: its pure scalar slice has all extra terms zero and is tested as a separate control; no existing tier is changed here.

6. Result footer

Result ID:	A1-PRODUCTION-FUNCTIONAL-REALISATION
Precise statement:	A reproducible audit freezes the declared production energy and tests its gradient/Hessian consistency.
Scope:	N=4 diagnostic grids; hash-pinned external working branch.
Dependencies:	A1-PRODUCTION-KERNEL-MANIFEST.
Evidence grade:	EXECUTED, CONDITIONAL.
Reproduction command:	python codes/foundations/a1_production_functional_realisation.py
Expected output:	independent energy control pass; scalar and full working-branch mismatches detected and recorded.
Falsification gate:	source hash drift, failed control, or a mismatch not detected.
Tier before / after:	none / T3 computable scaffold.
No-overclaim statement:	Not a PDE theorem, BCC result, or full closure.
Next required action:	choose and implement one Class-II functional convention, then rerun this audit before any scoped T5 review.